

Building a Retirement Income Treasury STRIPS Ladder

Building bond ladders for retirement income is an important but understudied topic. Especially since we are at a point when many are worried about future interest rate increases, bond mutual funds risk locking in capital losses when shares are sold to meet expenses, while a bond ladder will still provide the obligated cash flows, no matter what happens with interest rates. Rather than purchasing bond mutual funds, an individual could build a ladder of bonds with some maturing each year to provide the income needed to cover annual expenses.

The *Wall Street Journal*'s [Market Data Center](#) provides information on all outstanding Treasury bonds, STRIPS (zero-coupon bonds) and Treasury Inflation-Protected Securities (TIPS). Though creating an initial bond ladder with any of these asset subclasses is relatively straightforward, coupon payments increase the complexity, especially when it is time to extend the bond ladder as time passes. For this reason, we will first focus on US Treasury bond STRIPS.

Treasury STRIPS are zero-coupon bonds that only provide their face value on the maturity date. They sell for less than their face value, and the implied investment return from holding them to their maturity date can be calculated based on the pricing discount from their stated face value paid at maturity, as well as the length of time to maturity.

Using data from the *Wall Street Journal* for January 3, 2017, I've calculated the cost of building a ladder of Treasury STRIPS for up to thirty years, using the bond that matures at the earliest date each year (2017–2046). For the first year, cash is held for expenses rather than purchasing a bond that will mature in a few weeks. The results are in Exhibit 7.11.

Yields range from 0.56 percent for bonds maturing in 2018, up to 3.12 percent for bonds maturing in 2046. The bond ladder is designed to provide \$40,000 of income in the first year, with 2 percent annual growth to approximate inflation. Without TIPS, built-in cost-of-living adjustments can only be applied in such a way. Future realized inflation could differ from